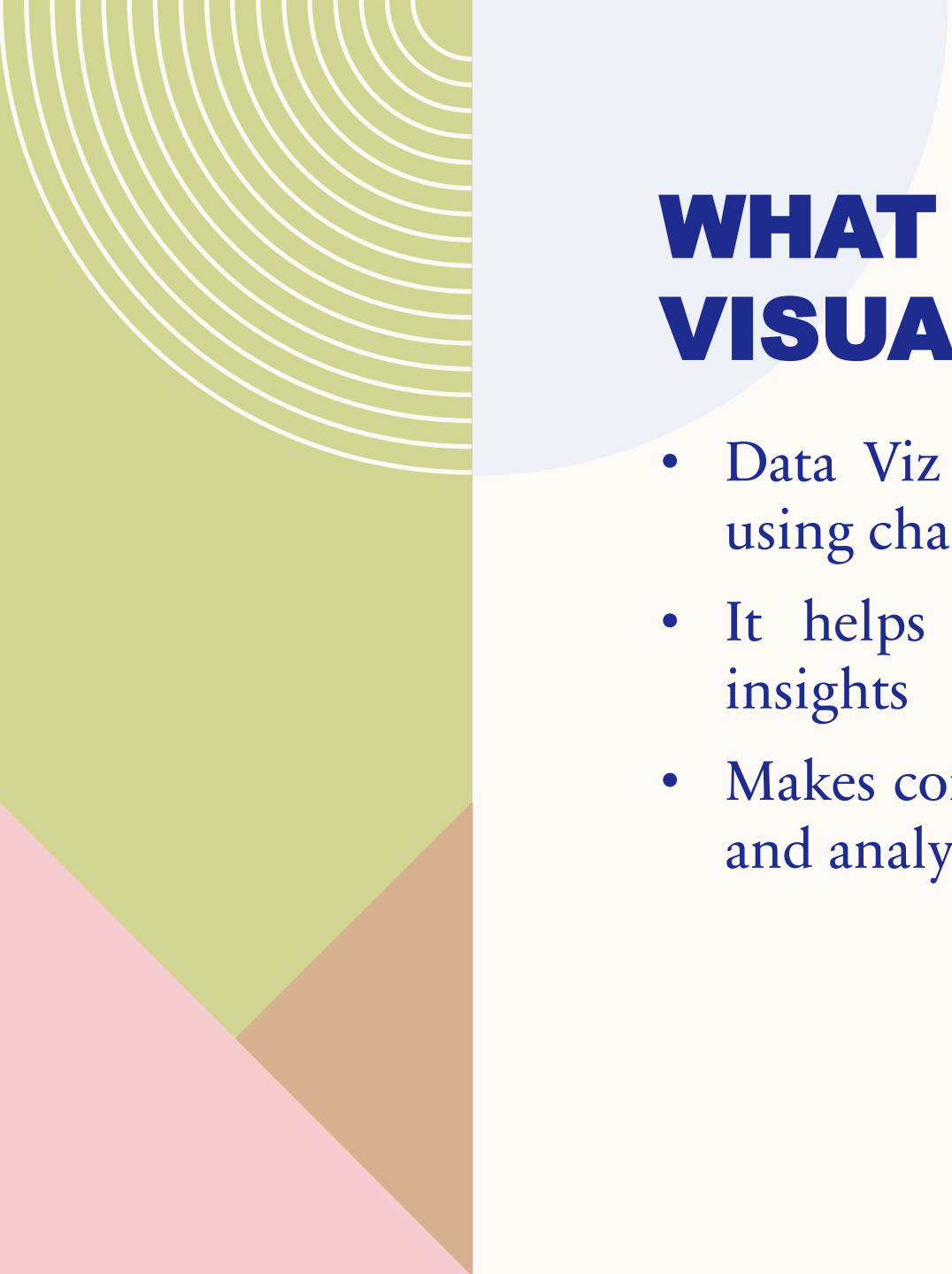


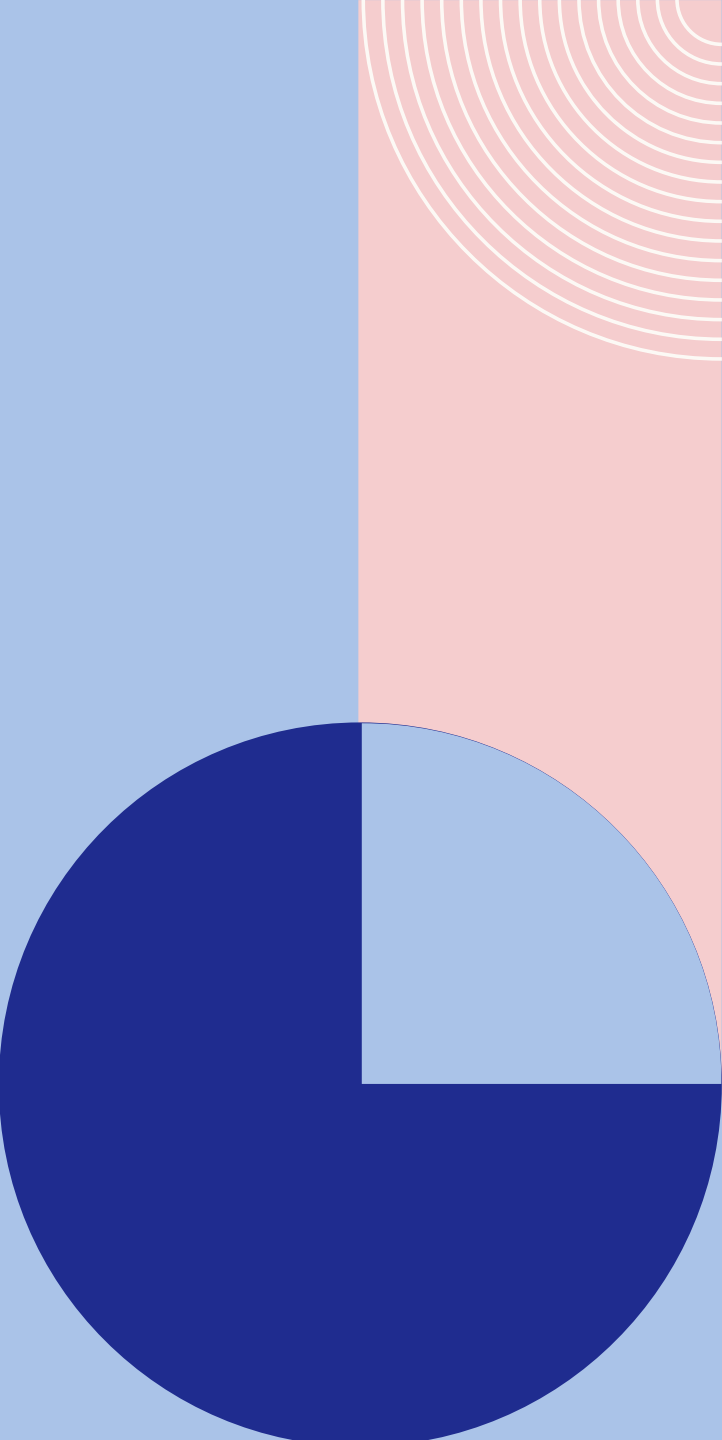
The background features a large white circle in the center, partially overlapping a light blue area on the left and a light pink area on the right. A dark blue shape is at the bottom, also overlapping the white circle. The text is centered within the white circle.

INTRO TO DATA VISUALIZATION



WHAT IS DATA VISUALIZATION?

- Data Viz is the graphical representation of data using charts, graphs, and visuals
- It helps transform raw data into meaningful insights
- Makes complex data easy to understand, compare, and analyze



WHY IT IS IMPORTANT?

- Helps to Identifies patterns, trends, and outliers quickly
- Helps in better decision-making
- Improves communication of data to non-technical users
- Simplifies complexity by converts massive datasets into simple digestible visuals
- Visuals allow quicker interpretation of data, helping spot key info quickly

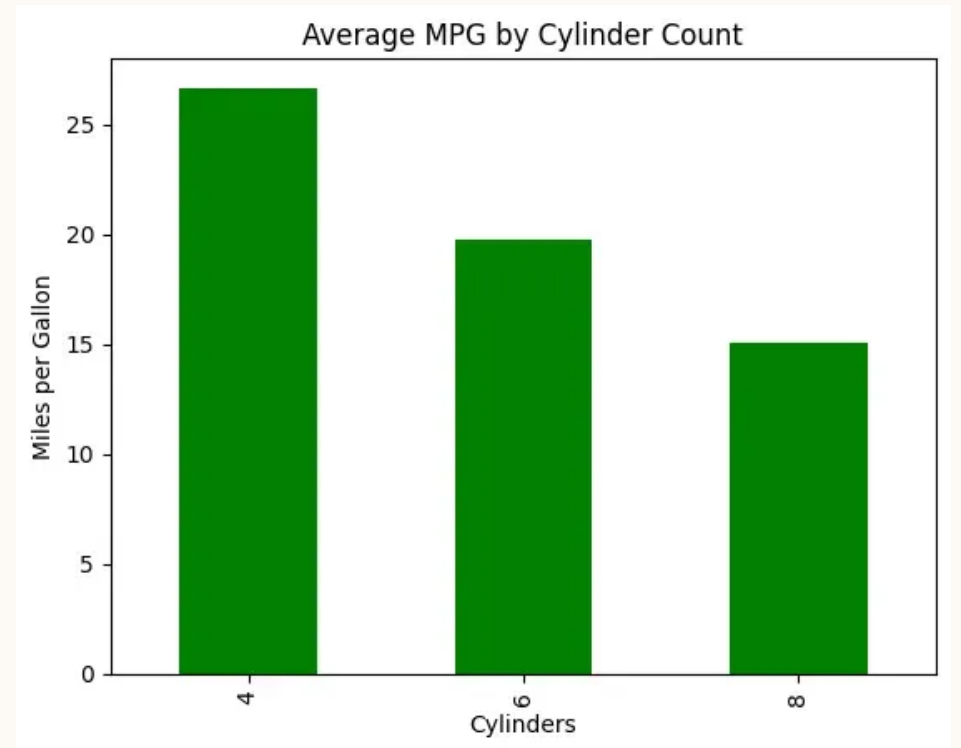
COMMON TYPES OF GRAPHS

Bar Charts

Bar Charts use rectangular bars to represent numerical values across different categories.

When to Use:

- To compare different categories
- To rank values from highest to lowest
- To show relationships between multiple variables

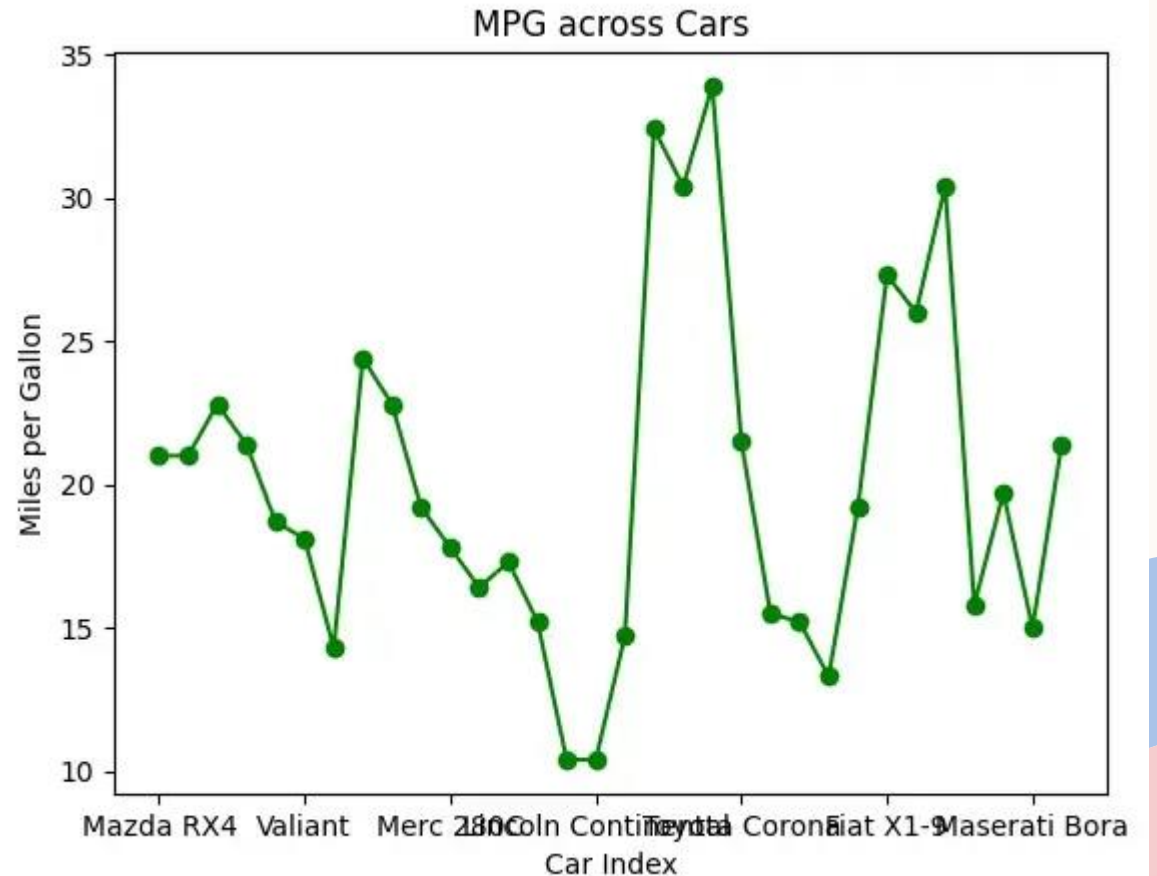


Line Charts

Line charts connect data points with lines across continuous intervals, such as time.

When to Use:

- To track changes over time
- To compare trends across multiple data series
- For time series analysis



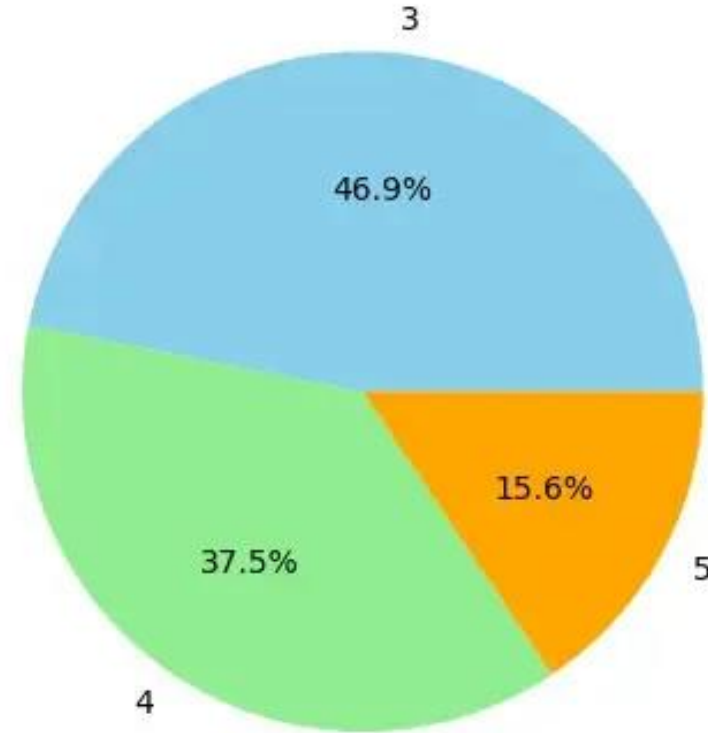
Pie Charts

Pie charts display data as proportional slices of a circle, representing each category's contribution to the whole.

When to Use:

- To show how different parts contribute to a whole
- To highlight a dominant category

Distribution of Cars by Gear Type

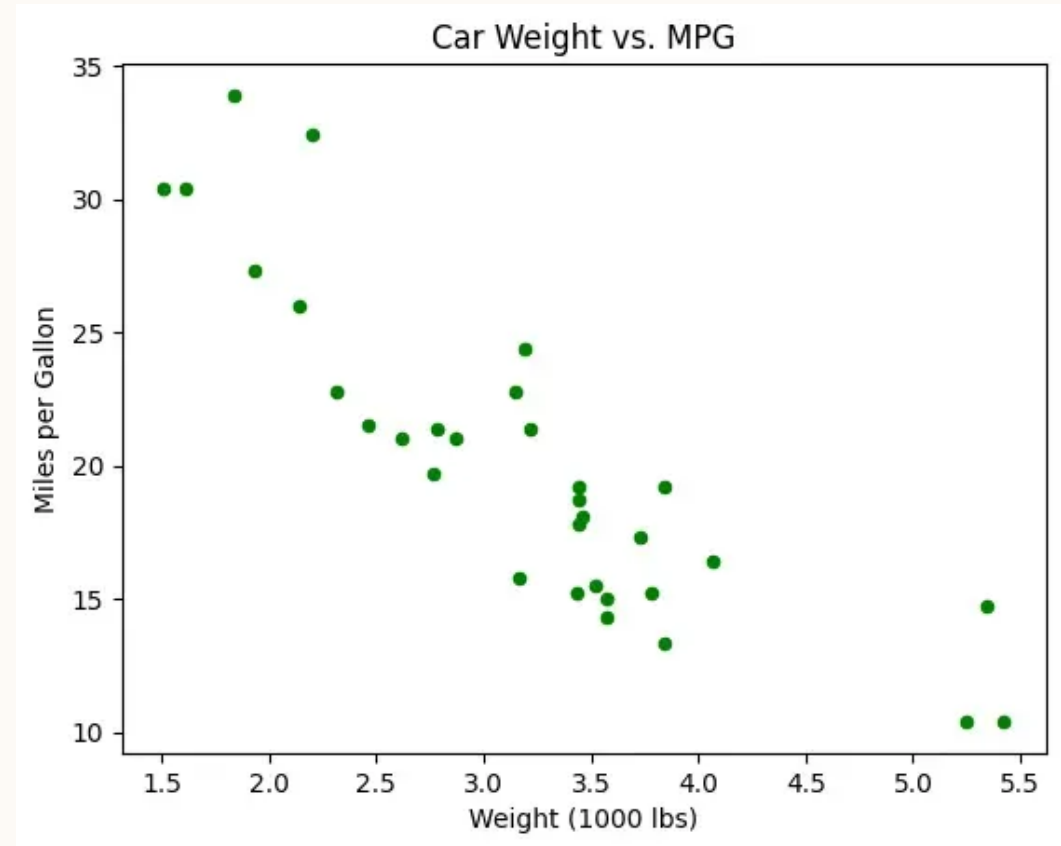


Scatter plots

Scatter plots represent relationships between two numerical variables using points on a 2D plane.

When to Use:

- To observe relationships between two variables
- To detect patterns, clusters or outliers in data

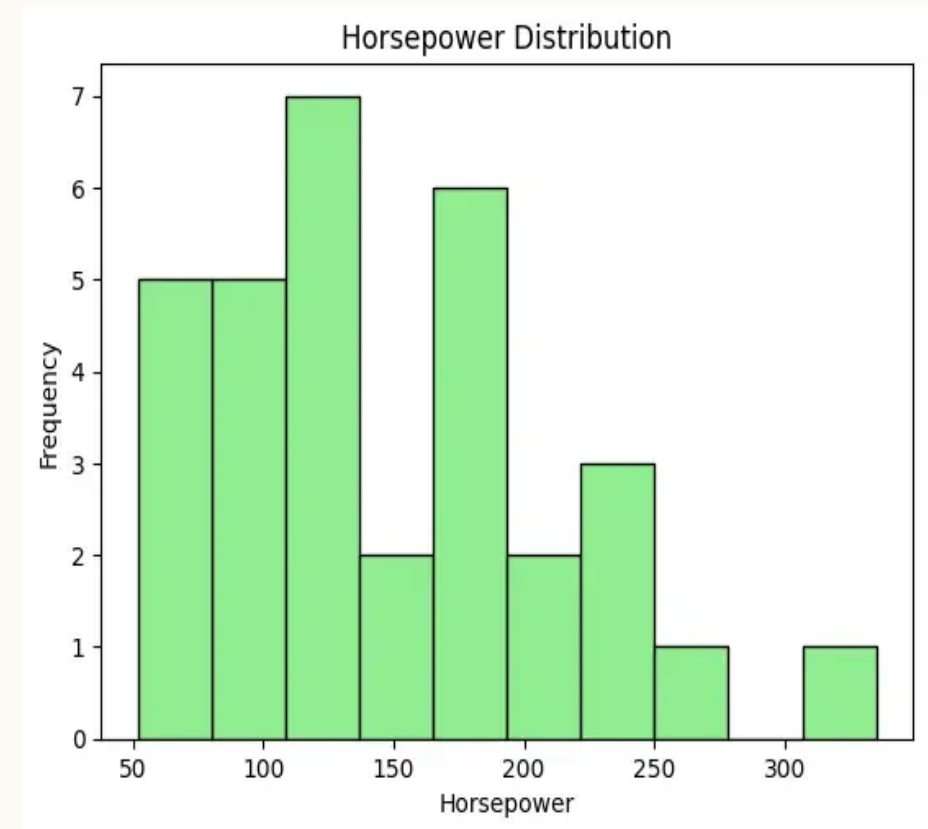


Histograms

Histograms display the distribution of continuous data by grouping values into intervals (bins).

When to Use:

- To visualize the distribution of numerical data
- To explore patterns, trends and outliers

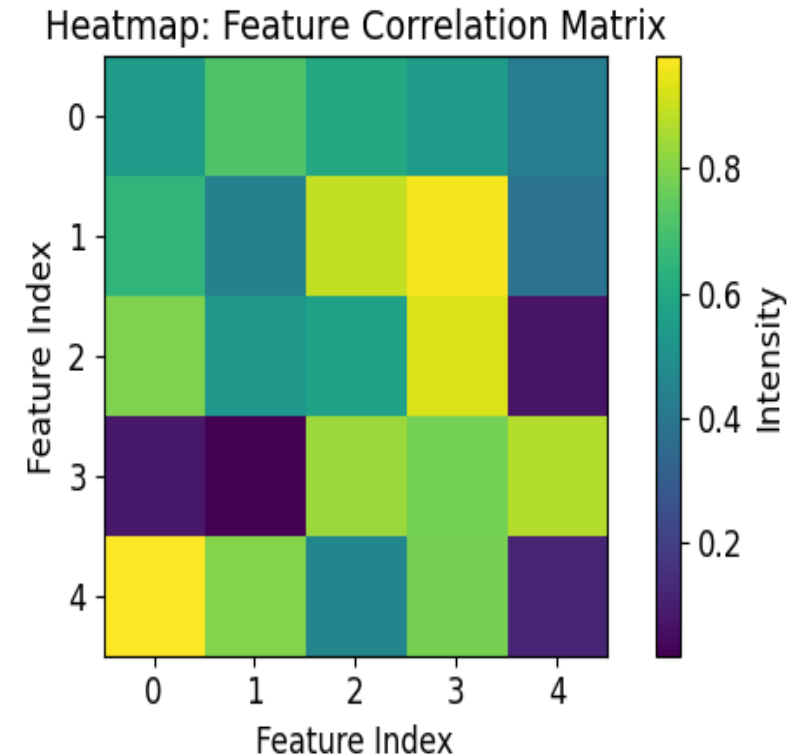


Heatmap

A heatmap displays data in a matrix format using color to represent values. It's ideal for spotting patterns, correlations and variations in large datasets.

When to Use:

- To identify clusters or groupings in data
- To visualize correlations between variables

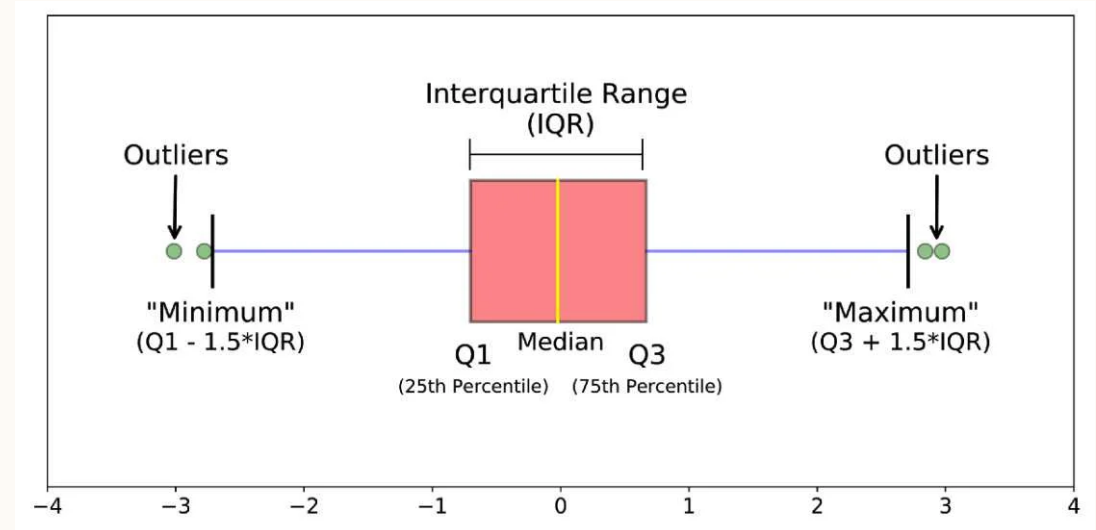


Box Plot

A box plot displays distribution of numerical data, showing median, quartiles and outliers. It's useful for understanding variability and detecting unusual values.

When to Use:

- To detect outliers in data
- To compare distributions across groups
- To visualize spread and variability



POPULAR DATA VISUALIZATION TOOLS

- Tableau: Known for powerful, interactive dashboards and mapping capabilities, good for complex data.
- Microsoft Power BI: Comprehensive analytics and visualization for businesses, integrates well with Microsoft ecosystem.
- Looker (Google Cloud): Premium, real-time data visualization and BI for businesses.



POPULAR DATA VISUALIZATION TOOLS

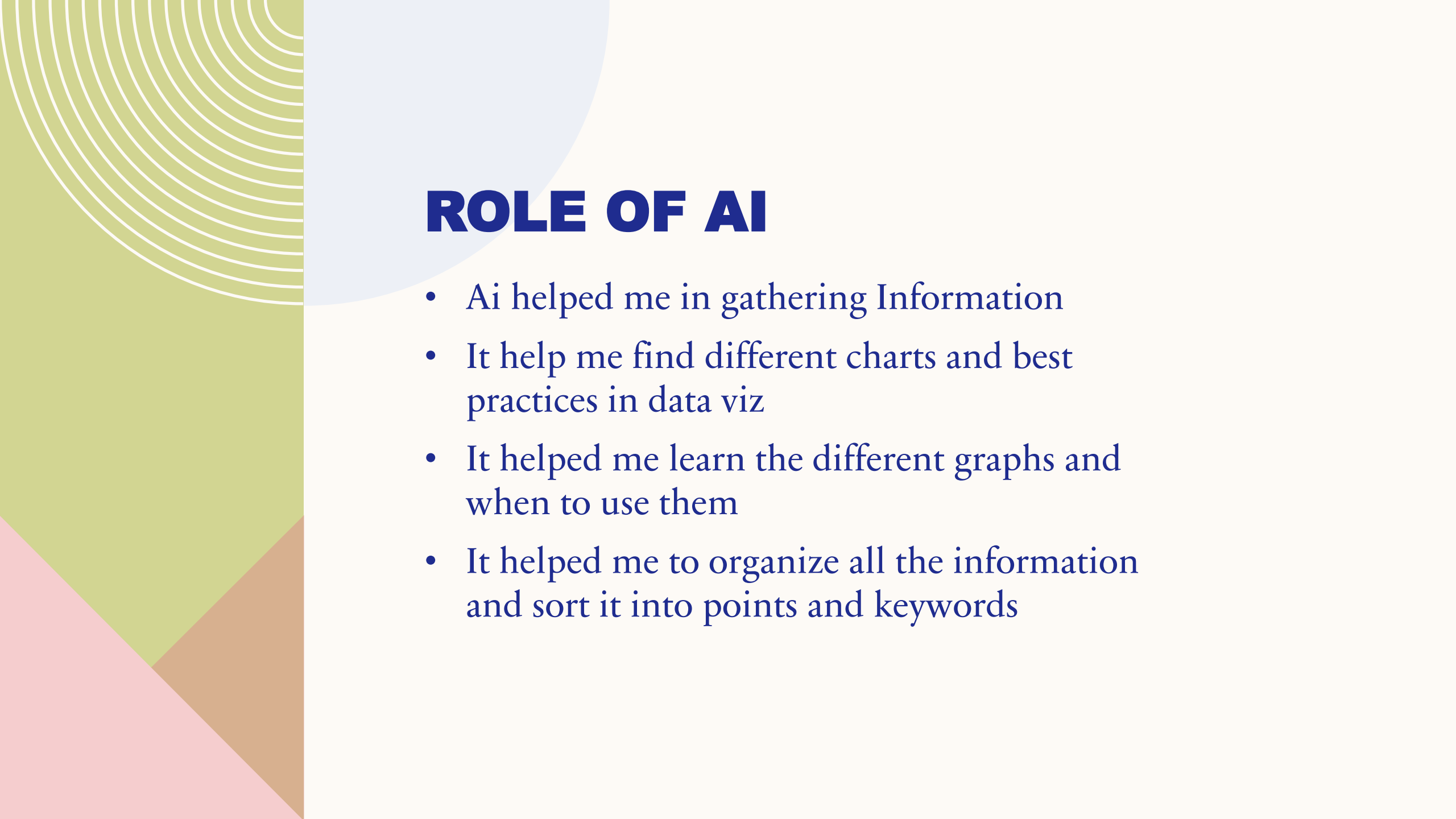
- Python libraries
 - Matplotlib: low-level, fully customizable
 - Seaborn: statistical and aesthetic plots
 - Plotly: interactive and web-ready visualizations
- Microsoft Excel: Basic charts, quick analysis, widely used





BEST PRACTICES

1. Choose the Right Chart
2. Keep Visuals Simple
3. Use Clear Titles and Labels
4. Maintain Consistent Colors
5. Use Appropriate Scales
6. Highlight Key Insights
7. Avoid Overcrowding
8. Use Readable Fonts
9. Ensure Data Integrity
10. Design for the Target Audience



ROLE OF AI

- Ai helped me in gathering Information
- It help me find different charts and best practices in data viz
- It helped me learn the different graphs and when to use them
- It helped me to organize all the information and sort it into points and keywords



THANK YOU